

Inequalities — 4

Inequalities for series and sequences

Homework

10 сентября 2026 года

Theoretical minimum. Abel summation and the bound it gives for a sum with monotone factors; the Dirichlet and Abel tests; Hardy's inequality with the sharp constant $(p/(p-1))^p$; Carleman's inequality with the constant e ; Hilbert's inequality with the constant π ; the common proof scheme — split the term, apply Hölder with a weight, optimise the weight.

1. Prove Hilbert's inequality with the constant π , and show that the same scheme gives the integral analogue $\iint \frac{f(x)g(y)}{x+y} dx dy \leq \pi \|f\|_2 \|g\|_2$.
2. (Putnam 2014) Assume that $(a_n)_{n \geq 1}$ is an increasing sequence of positive reals with $\lim a_n/n = 0$. Must there exist infinitely many n such that $a_{n-i} + a_{n+i} < 2a_n$ for $i = 1, 2, \dots, n-1$?
3. (Proposed problems) Let (x_n) be a sequence of reals with $\limsup_{n \rightarrow \infty} (1 - x_n) \log n < \infty$. Show that if a series $\sum a_n$ with $a_n > 0$ converges, then $\sum a_n^{x_n}$ converges as well.
4. (Proposed problems) Let $(b_n)_{n \geq 0}$ be positive. Prove that the following are equivalent: $\sum \frac{|b_{n+1} - b_n|}{b_n} < \infty$; $\sum \frac{|b_{n+1} - b_n|}{b_{n+1}} < \infty$; and $\sum |b_{n+1} - b_n| < \infty$ together with $\lim b_n > 0$.
5. (Proposed problems) Let p_n be the n -th prime number. Show that the sequence $x_n = \left(\frac{1}{p_1} + \frac{1}{p_2} + \dots + \frac{1}{p_n}\right) - \{\log \log n\}$ is divergent, where $\{t\}$ denotes the fractional part of t .
6. (Putnam 2016) Prove that for every set $X = \{x_1, \dots, x_n\}$ of n real numbers there is a non-empty subset S of X and an integer m with $|m + \sum_{s \in S} s| \leq \frac{1}{n+1}$.

7. (Putnam 2015) Let $\mathcal{C}$ be the set of cosine polynomials $f(x) = 1 + \sum_{n=1}^N a_n \cos(2\pi nx)$ such that $f \geq 0$ everywhere and $a_n = 0$ whenever $3 \nmid n$. Determine the maximum value of $f(0)$ over $\mathcal{C}$ and prove that it is attained.
8. Prove the discrete Chebyshev sum inequality from Abel summation, and use Abel summation to prove that $|\sum_{k=1}^n \frac{\sin kx}{k}|$ is bounded by a constant independent of n and x .
9. Prove Knopp's inequality $\sum_{n \geq 1} (a_1 a_2 \cdots a_n)^{1/n} \leq e \sum_{n \geq 1} a_n$ directly, without passing to the limit in Hardy's inequality, by using the weights $w_k = \frac{(k+1)^k}{k^{k+1}}$ and the arithmetic-geometric mean inequality.
10. Let (a_n) be positive with $\sum a_n$ divergent. Prove that $\sum \frac{a_n}{a_1 + \cdots + a_n}$ also diverges, while $\sum \frac{a_n}{(a_1 + \cdots + a_n)^2}$ converges.